

Software Requirement Engineering

Project Name: Solar Quotation Management System

Submitted By:

- *Bsef23m544 Waleed Daud*
- *Bsef23m509 M. Adil Nadeem*
- *Bsef23m540 Ahmad Bilal*
- *Bsef23m528 M. Sheheryar*

Submitted To:

- *Dr. Amina Mustansir*



DEPARTMENT OF SOFTWARE ENGINEERING, PUNJAB UNIVERSITY, LAHORE

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

The purpose of the Solar Quotation Management System (SQMS) is to automate and streamline the process of generating, managing, and tracking solar installation quotations for customers. The system aims to improve efficiency, reduce manual efforts, and ensure accurate tracking of quotations, customer data, and employee performance.

1.2 Scope

The system will provide features for employees to create quotations, manage customer information, and track performance and sales. It will also allow administrators to configure the system, set rates, and track the performance of employees. The system will replace existing manual processes, such as using Excel sheets and templates, and will be a web-based platform accessible by employees and admins.

1.3 Definitions, Acronyms, and Abbreviations

- **SQMS:** Solar Quotation Management System
- **Admin:** Administrator of the system, responsible for configurations and employee management.
- **Employee:** Users who generate quotations and manage customer interactions.
- **Customer:** Individuals seeking solar installation quotes and information.
- **ROI:** Return on Investment

1.4 References

- IEEE 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

1.5 Overview

This document presents the detailed software requirements for the Solar Quotation Management System (SQMS). It covers the functional and non-functional requirements of the system, along with the user roles, database structure, and quality attributes necessary for the system's implementation.

2. Overall Description

2.1 Product Perspective

The Solar Quotation Management System (SQMS) is a web-based platform that will enable employees to create and manage solar installation quotations, track customer interactions, and manage the performance of the sales team. The system will replace existing manual processes and will provide a streamlined workflow for generating quotations, managing data, and communicating with customers.

2.2 Features

- **Quotation Creation:** Employees can create initial and detailed quotations for solar installations, incorporating customer data, site requirements, and solar panel details.
- **Rate Management:** Admins can set and update rates, while employees can use predefined rates or enter custom values for quotations.
- **Customer Management:** Customers' information, including contact details and interaction history, will be stored in the system for future reference and follow-ups.
- **Performance Tracking:** Admins can track employee performance, including sales and commissions, and generate reports for analysis.
- **Expense Management:** The system will include features to manage and track expenses related to quotations, such as solar panels, tools, transportation, and site visit costs.
- **Commission Tracking:** Admins can monitor employee commissions based on performance and sales.

2.3 User Characteristics

- **Admin:** Has full access to all system configurations, employee management, and performance reports. Responsible for setting rates, managing users, and configuring the system.
- **Employee:** Can create quotations, manage customer interactions, and view their own performance data. Employees will have access to predefined data or can enter custom values for quotations.
- **Customer:** Interacts with the system indirectly by receiving quotations and reports created by employees. Customers' contact information and quotation history will be stored in the system.

2.4 Constraints

- The system will be web-based, requiring internet access for employees and admins.
- The system must be user-friendly for employees with varying technical expertise.
- Data security and privacy must be ensured, particularly for customer information.
- Performance must support up to 500 concurrent users.

2.5 Assumptions and Dependencies

- Employees will have internet access and compatible devices to use the web-based system.
- The system will be accessible via modern web browsers (Chrome, Firefox, etc.).
- The system assumes that users will have basic knowledge of managing quotations and customer interactions.
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3. Specific Requirements

3.1 External Requirements

- **User Interfaces:** The system will provide a web-based interface accessible via modern browsers (Chrome, Firefox, etc.).

- **Hardware Interfaces:** Standard devices such as laptops, tablets, and smartphones will be used by employees to access the system.
- **Communications Interfaces:** The system will communicate over HTTP/HTTPS, with email notifications for quotations and updates to customers.

3.2 Logical Database Requirements

- **Customer Data:** The database will store customer details, including contact information and quotation history.
- **Employee Data:** The database will store employee information such as login credentials, performance data, and commission details.
- **Quotation Data:** Quotations will be stored with details such as panel types, quantities, customer data, rates, expenses, and ROI.
- **Expense Data:** The system will store data related to expenses for each quotation, such as panel costs, transportation, and site visits.

3.3 Design Constraints

- The system must be designed to support easy scalability to accommodate a growing number of users and quotations.
- The system must ensure secure storage and transfer of sensitive data, including customer and employee information.
- The system should be developed using a modern web framework (e.g., React or Angular for frontend and Node.js or Django for backend).

3.4 Software System Quality Attributes

- **Performance:** The system should be able to handle up to 500 concurrent users with minimal latency.
- **Security:** User data (including customer information) must be encrypted and securely stored. Access to sensitive data should be role-based.

- **Usability:** The system should be easy to use for employees with varying levels of technical expertise.
 - **Maintainability:** The system should be modular and easy to maintain with well-documented code.
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4. Use Case Descriptions

4.1 UC1: Log in to the System

Use Case Description

- **Scope:** The authentication system of the application.
- **Level:** User-goal.
- **Primary Actor:** User (employee or administrator).

Stakeholders and Interests

- **User:** Wants to securely log in and access role-specific functionality.
- **System Administrator:** Ensures secure login to prevent unauthorized access.
- **Organization:** Requires secure authentication to protect user data and ensure system integrity.

Preconditions

- The user must have valid credentials (username and password).
- The system must be available and accessible.
- The user account should not be locked or disabled.

Success Guarantee

The user is authenticated and redirected to their role-specific dashboard (employee or administrator).

Main Success Scenario

1. The user navigates to the login page.
2. The user enters their valid username and password.
3. The user clicks the "Login" button.
4. The system validates the entered credentials.
5. If valid, the system terminates any active sessions for the same user (if applicable).
6. The system identifies the user's role (employee or administrator).
7. The system redirects the user to their role-specific dashboard.

Extensions

1a. **Invalid Credentials:** The system displays an error message: "Invalid username or password."
1b. **Empty Fields:** The system displays inline error messages: "This field cannot be empty."
1c. **Account Lockout:** After 5 failed login attempts, the system locks the user account temporarily and displays: "Too many failed attempts. Please try again later."
3a. **Password Reset:** The user clicks the "Forgot Password" link. The system redirects the user to the password recovery page.
4a. **Password Visibility Toggle:** The user clicks the "eye" icon to toggle password visibility.

Special Requirements

- Passwords must meet the system's security requirements (e.g., a minimum of 8 characters).
- The system must securely store and validate credentials using encryption.
- The system should terminate any existing active sessions after successful login.

4.2 UC2: Add an Employee

Use Case Description

- **Scope:** Employee Management System.
- **Level:** User-goal.
- **Primary Actor:** Administrator.

Stakeholders and Interests

- **Administrator:** Efficiently add new employees with accurate data.
- **HR Department:** Ensures employee information is recorded for payroll, insurance, and other employee benefits.
- **Company:** Ensures employee data is accurately recorded, validated, and stored for operational efficiency and compliance.
- **System:** Handles the data securely and ensures all validation and data processing requirements are met.

Preconditions

- Administrator is logged into the system.
- Administrator has appropriate permissions to manage employee records.

Success Guarantee

A new employee record is added to the system's database. Employee details are validated, and no duplicate records are created.

Main Success Scenario

1. The Administrator navigates to the "Manage Employees" page.
2. The Administrator clicks the "Add Employee" button.
3. The system redirects the Administrator to the Add Employee Form.

4. The Administrator enters all required employee details (e.g., name, email, contact number, department).
5. The system validates the input fields for accuracy, including checking for duplicates in the email and contact number.
6. If all fields are valid, the Administrator clicks the "Save" button.
7. The system stores the employee data in the database.
8. The system displays a confirmation message: "Employee added successfully."

Extensions

5a. **Validation Errors:** The system highlights invalid fields (e.g., "Invalid email format"). 6a. **Duplicate Record Found:** The system displays an error message: "An employee with this email/contact number already exists." 7a. **Cancel Operation:** If the Administrator clicks "Cancel" or navigates away from the form, the system discards all unsaved data.

Special Requirements

- The system should validate email format and ensure contact numbers are numeric and meet length criteria.
- Duplicate prevention is required to ensure no two employees share the same email or contact number.
- Employee data should be stored securely in the system database.

4.3 UC3: Set and Configure Rates

Use Case Description

- **Scope:** Admin Dashboard.
- **Level:** User-goal.
- **Primary Actor:** Admin.

Stakeholders and Interests

- **Admin:** Manages rates and configurations efficiently.
- **End Users:** Depend on the system's accurate pricing for their transactions.
- **System:** Ensures that rates are stored and updated correctly, with appropriate validation.
- **Business:** Ensures proper business rules for rate configuration are followed to avoid pricing conflicts.

Preconditions

- The admin must be logged into the system.
- The admin must have the necessary permissions for rate management.
- The system must be running and able to interact with the database for saving rate configurations.

Success Guarantee

Rates and configurations are saved successfully in the database. The system displays a confirmation message, ensuring the admin that the rates have been updated.

Main Success Scenario

1. Admin navigates to the "Settings" menu in the dashboard.
2. Admin selects "Rates and Configuration" from the menu.

3. The system displays the configuration form for inputting rate details.
4. Admin enters the rate type, rate value, and applicable criteria (e.g., region, time, or user category).
5. The system validates the input fields for accuracy and completeness.
6. Admin clicks "Save," and the system stores the rates in the database.
7. The system displays a success message: "Rates and configurations have been successfully updated."

Extensions

- **Input Validation Fails:** If the rate value is not a positive number, an error message is displayed: "Rate value must be a positive number."
- **Unsaved Changes Warning:** If the admin attempts to navigate away from the page without saving changes, the system shows a confirmation: "You have unsaved changes. Do you want to discard them?"

4.4 UC4: Enter Customer Data for Quotation

Use Case Description

- **Scope:** Solar House Quotation Management System.
- **Level:** User-goal.
- **Primary Actor:** Salesperson.

Stakeholders and Interests

- **Salesperson:** Inputs accurate customer data for quotations.
- **Customer:** Expects accurate and timely quotations.

- **System Administrator:** Ensures data integrity and compliance.

Preconditions

- Salesperson is logged into the system.
- The dashboard is accessible with the "Create Quotation" section available.

Success Guarantee

Customer data is validated, saved securely, and linked to a new or existing quotation.

Main Success Scenario

1. The Salesperson navigates to the "Create Quotation" section.
2. The system displays a form to input customer details (name, contact, address, energy requirements).
3. The Salesperson fills in all mandatory fields with valid data.
4. The system validates the input (e.g., email format, phone number format).
5. The Salesperson clicks the "Save" button.
6. The system saves the data to the database, associates it with a quotation, and displays a success message.

Extensions

1a. **Invalid Input:** The system highlights invalid fields with error messages (e.g., "Invalid email format"). 2a. **Mandatory Fields Missing:** The "Save" button remains inactive until all mandatory fields are filled.

5. Specific Requirements

5.1 External Requirements

- **User Interfaces:** The system will provide a web-based interface accessible via modern browsers (Chrome, Firefox, etc.).

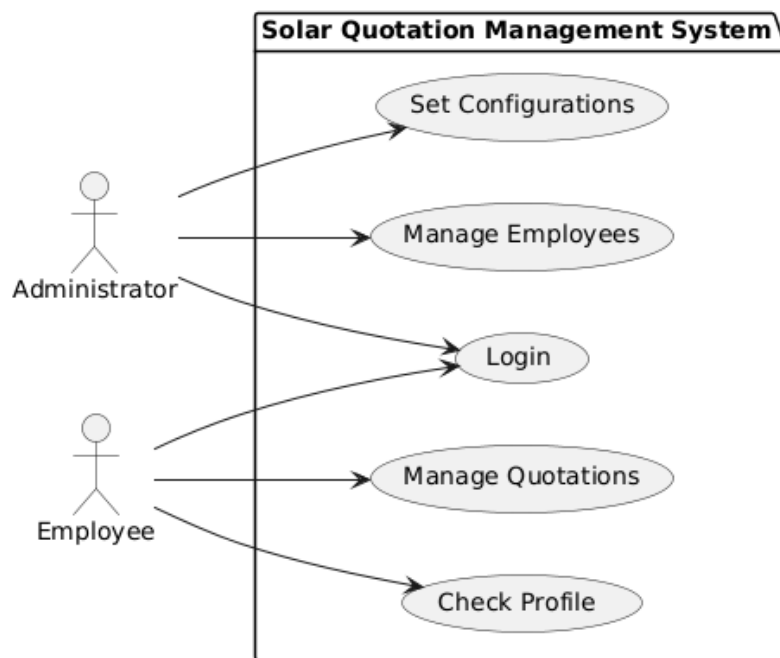
- **Hardware Interfaces:** Standard devices such as laptops, tablets, and smartphones will be used by employees to access the system.
- **Communications Interfaces:** The system will communicate over HTTP/HTTPS, with email notifications for quotations and updates to customers.

5.2 Logical Database Requirements

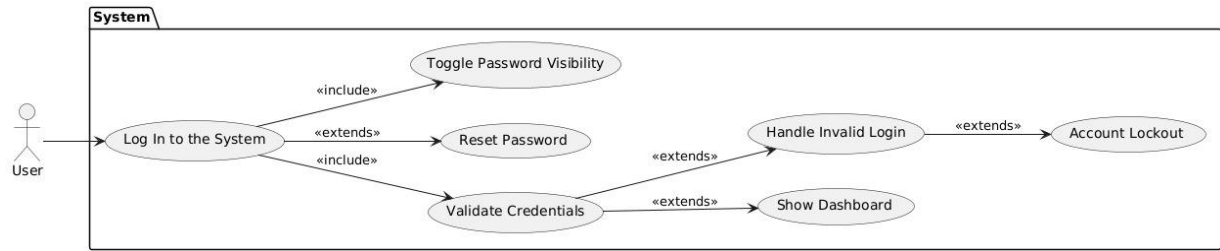
- **Customer Data:** The database will store customer details, including contact information and quotation history.
- **Employee Data:** The database will store employee information such as login credentials, performance data, and commission details.
- **Quotation Data:** Quotations will be stored with details such as panel types, quantities, customer data, rates, expenses, and ROI.

6. Appendices

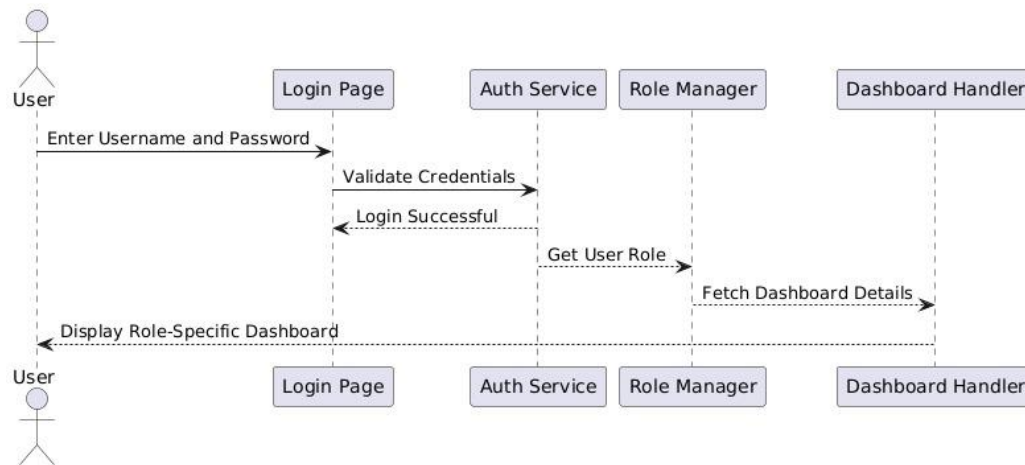
HIGH LEVEL USE CASE DIAGRAM



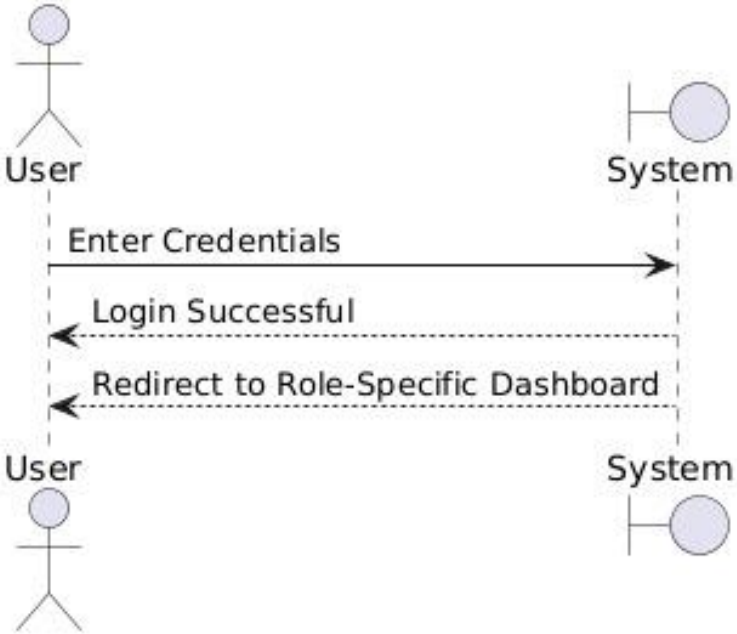
UC ANALYSIS DIAGRAM: LOGIN



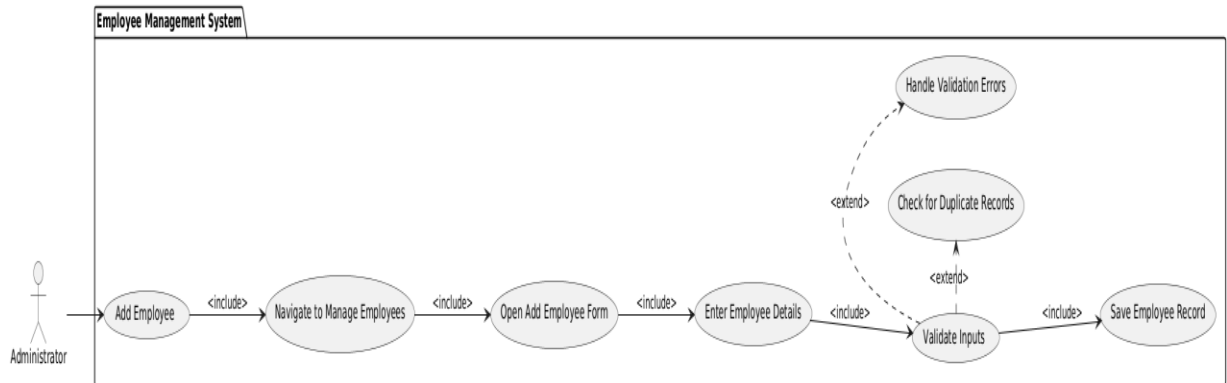
SD SCENARIO: LOGIN



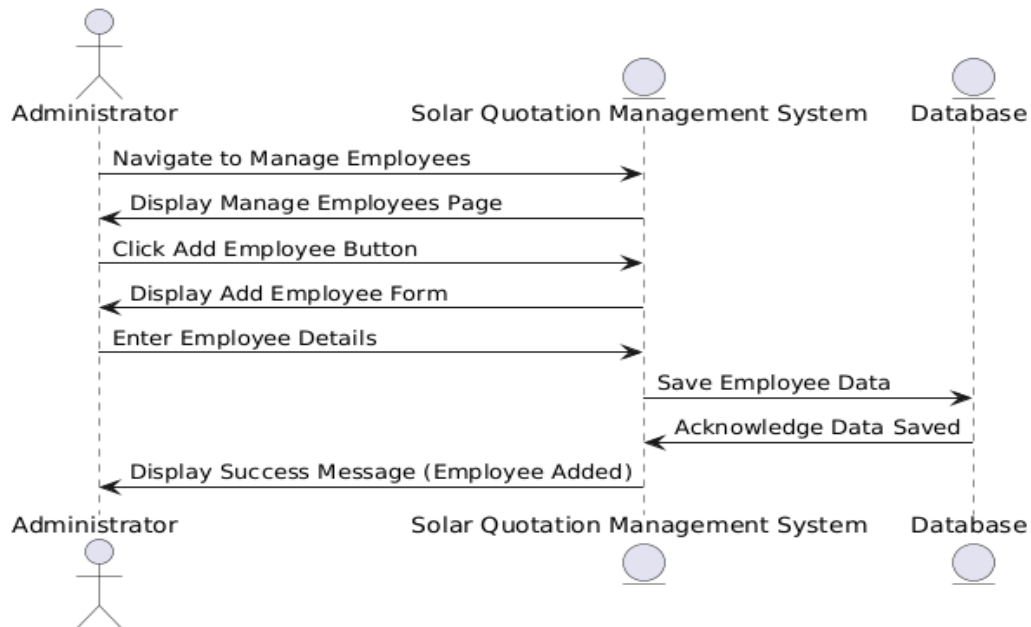
SSD SCENARIO LOGIN



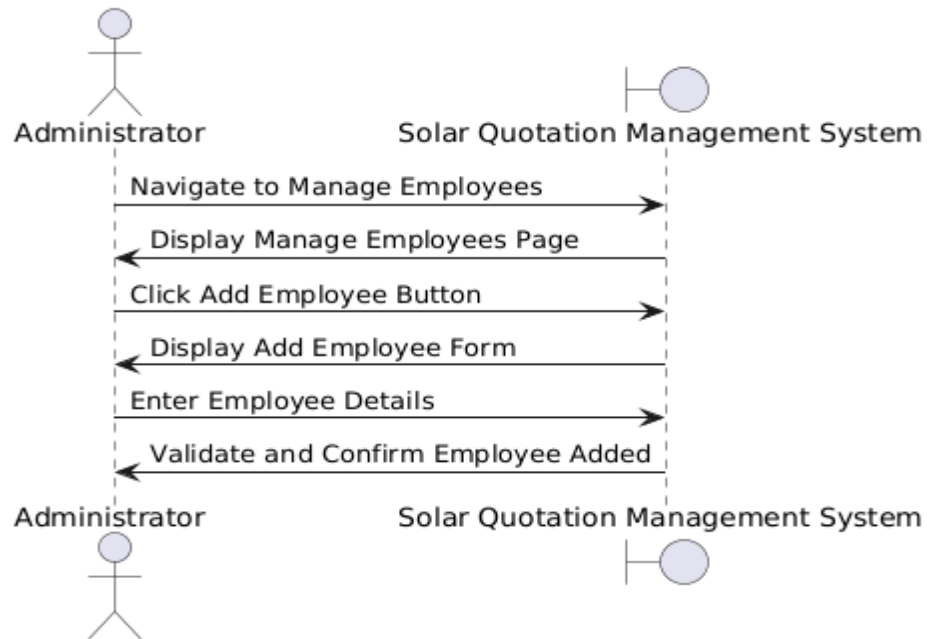
Analysis Level Diagram



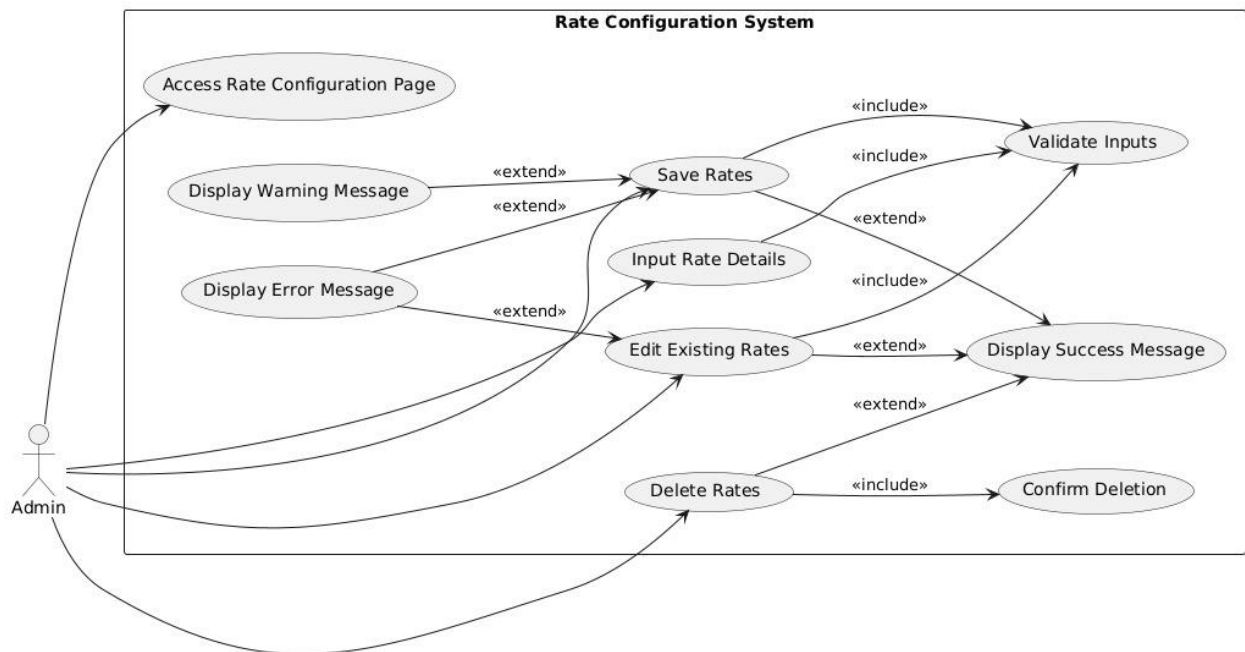
SD SCENARIO: ADD AN EMPLOYEE



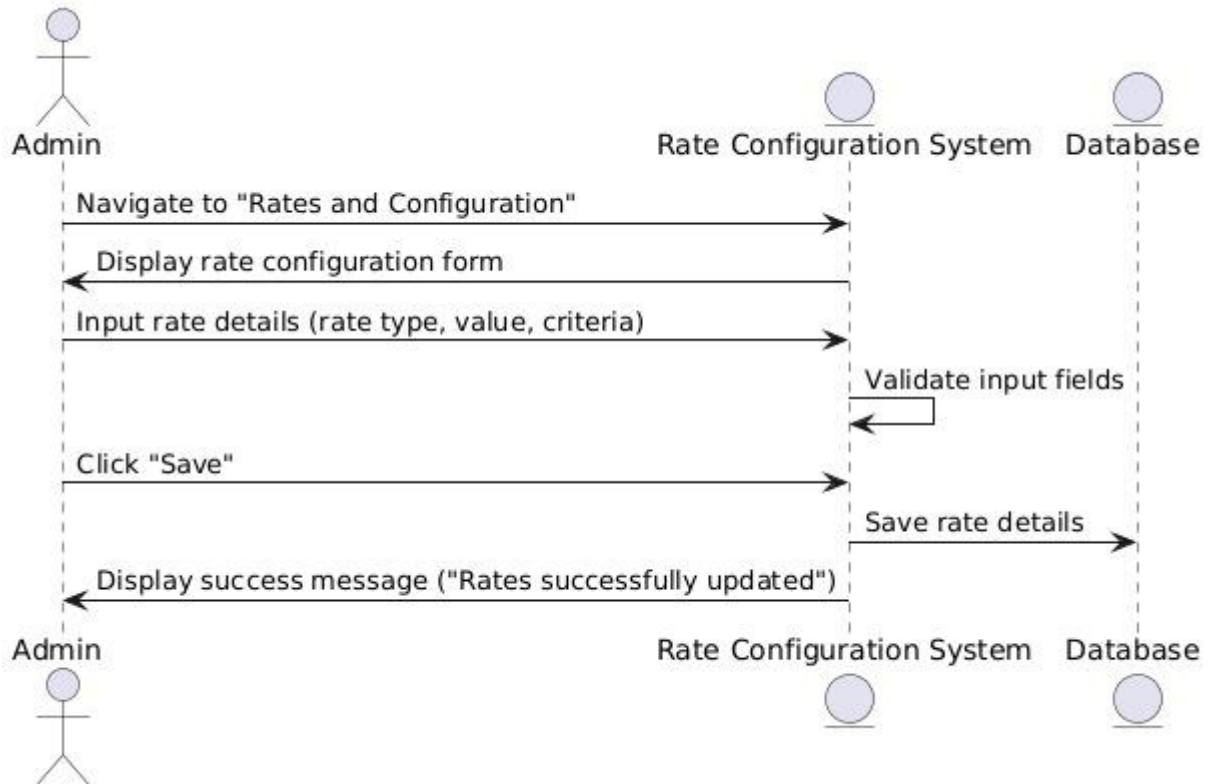
SSD SCENARIO: ADD AN EMPLOYEE



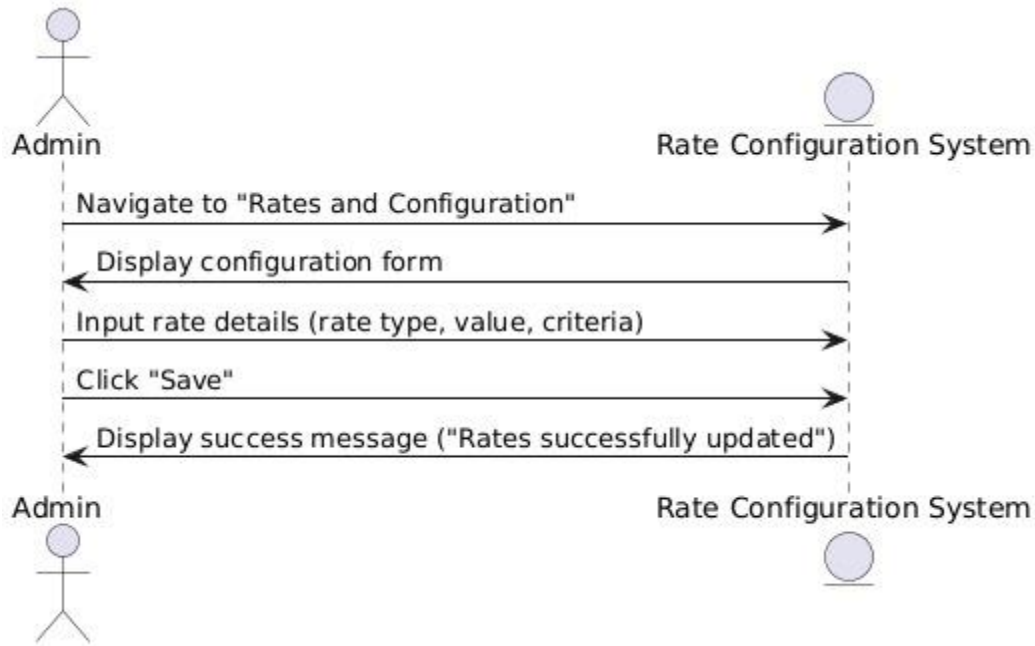
USECASE SET AND CONFIGURE RATES USECASE DIAGRAM



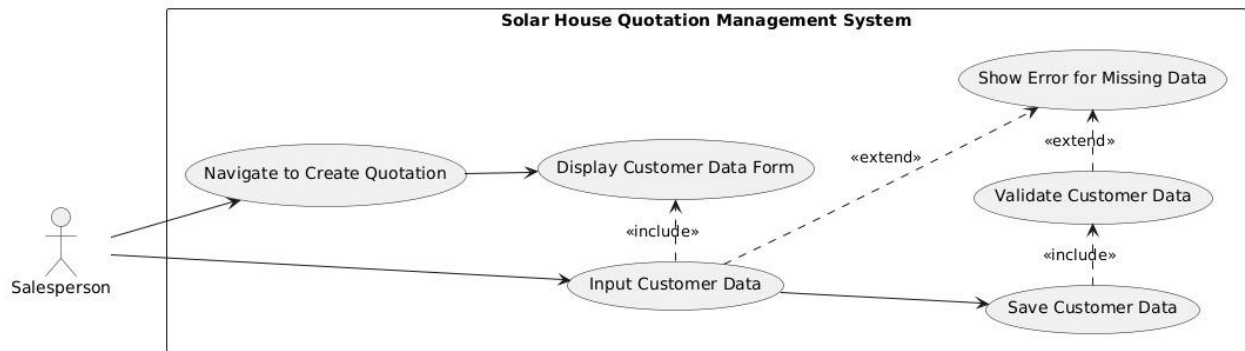
SD SCENARIO: SET AND CONFIGURE RATES



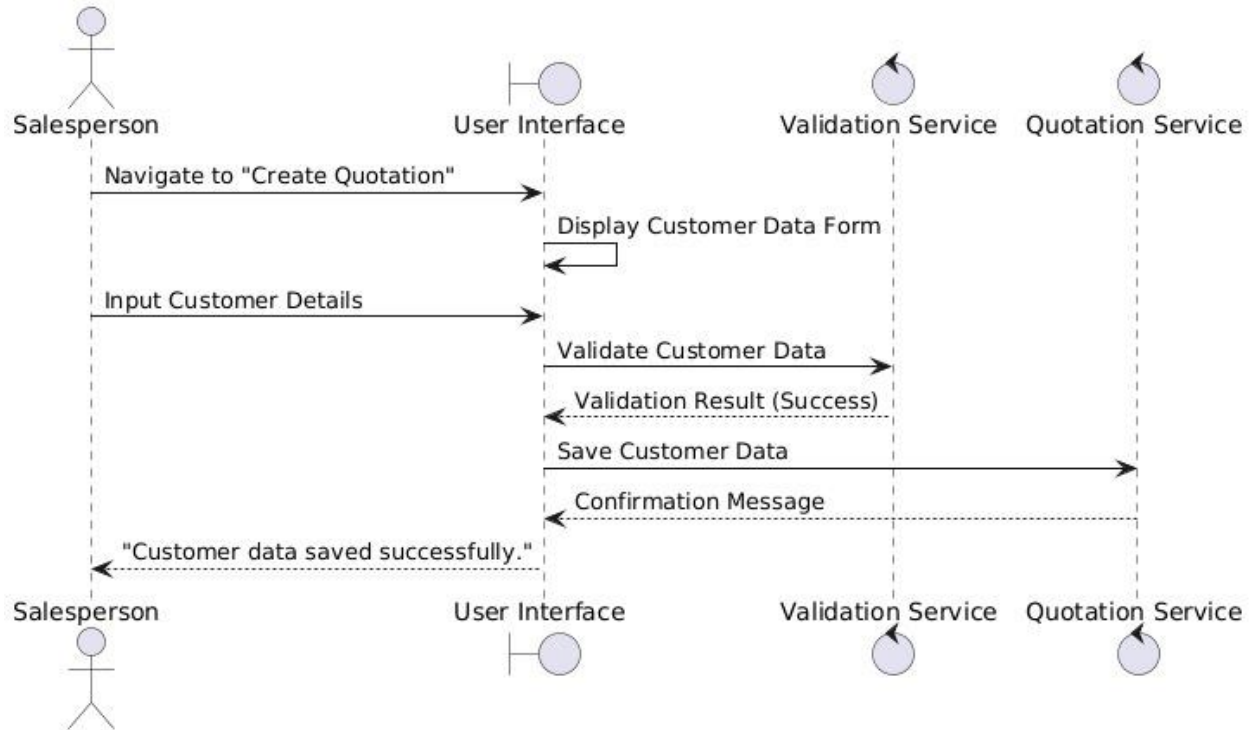
SSD SCENARIO: SET AND CONFIGURE RATES



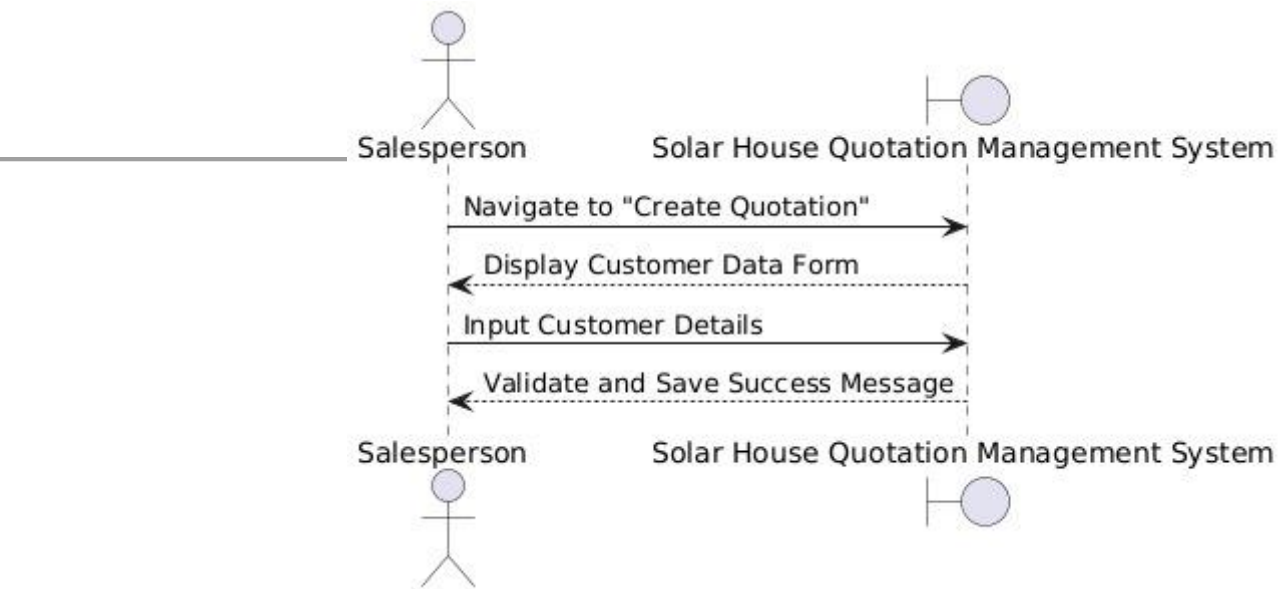
UC4: Enter Customer Data for Quotation Use case Diagram



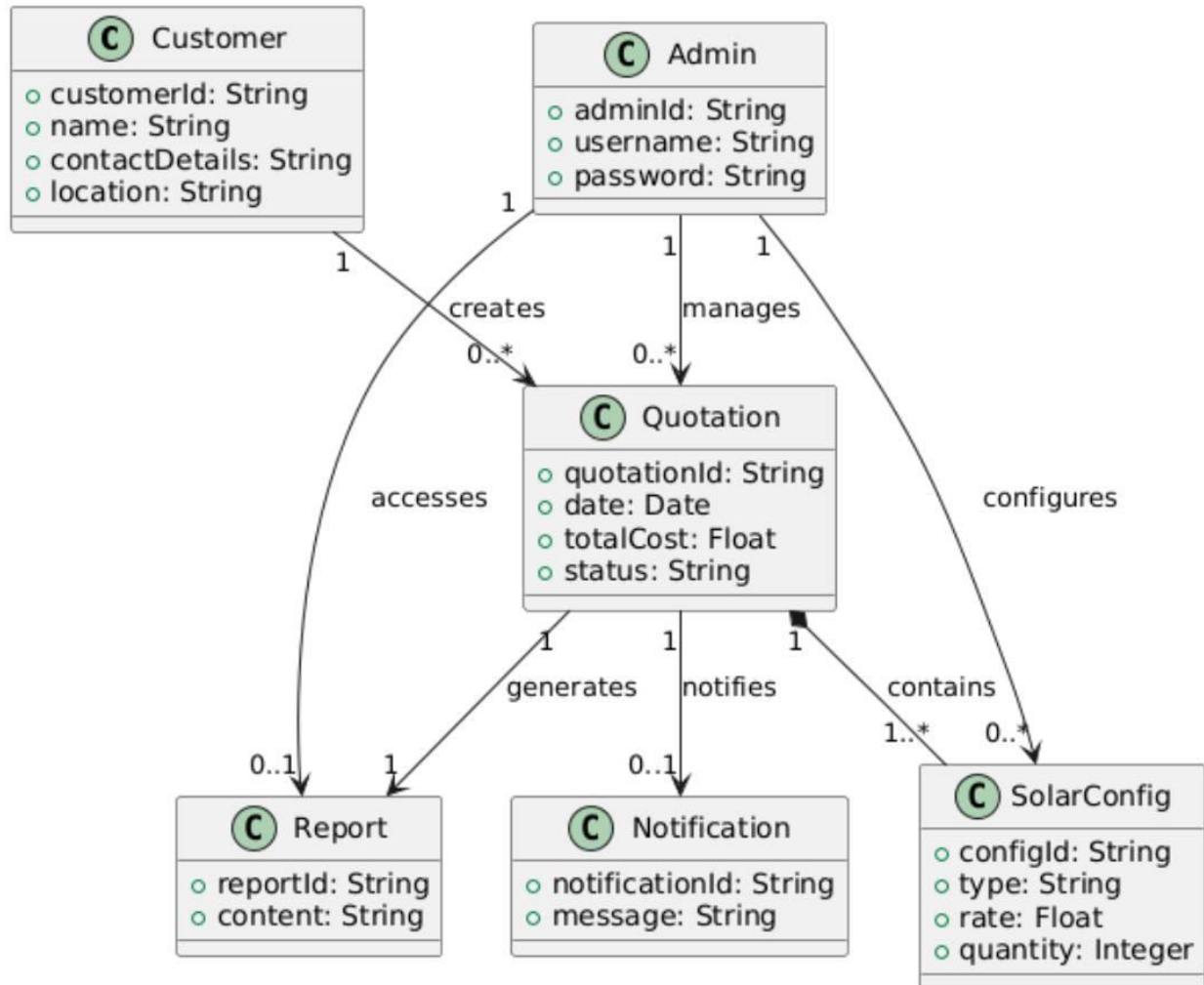
SD SCENARIO: CREATE QUOTATION



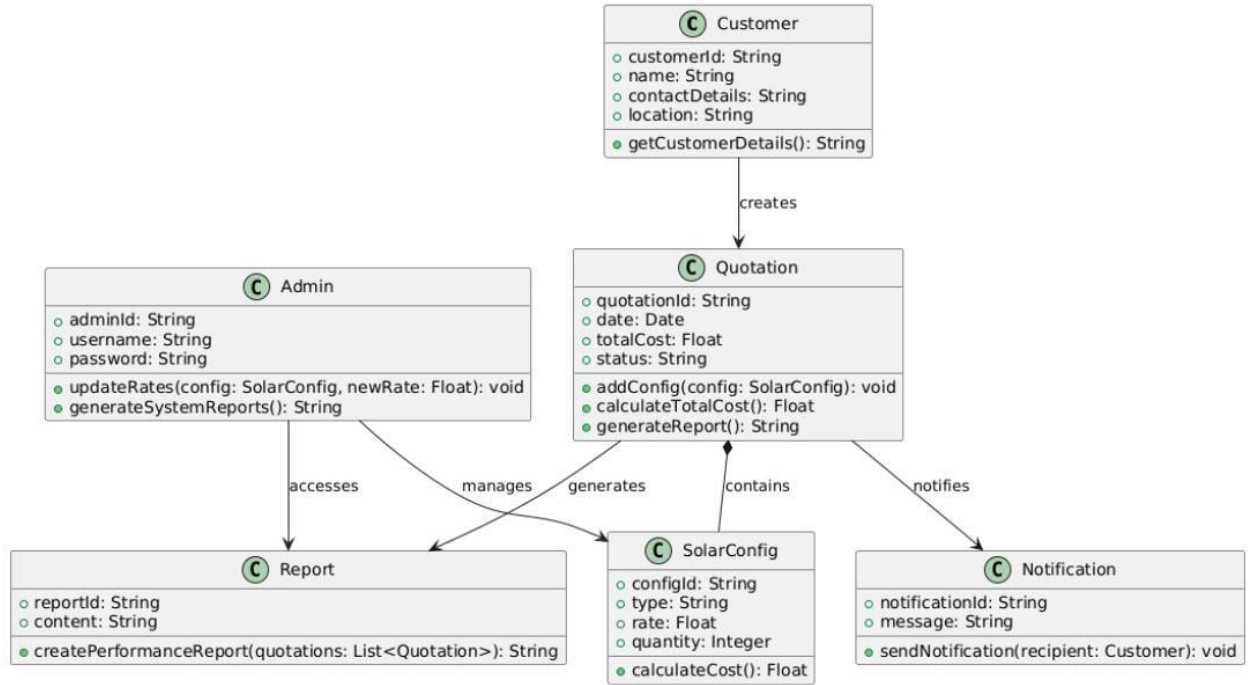
SSD SCENARIO: CREATE QUOTATION



CLASS DIAGRAM



UNIFIED MODELLING LANGUAGE



Glossary

1. **Admin:** A user role with full access to the system's configuration, management of employees, and access to performance data. Admins can set rates, update system configurations, and generate reports.
2. **Employee:** A user role responsible for generating and managing solar installation quotations for customers. Employees can interact with customer data, create quotations, and track customer interactions.
3. **Customer:** An individual or organization seeking a quotation for solar installation services. Their information, including contact details and quotation history, is stored in the system for follow-up and analysis.
4. **Quotation:** A formal offer generated by employees for solar installation services, which includes detailed information about the panels, installation requirements, costs, benefits, and return on investment (ROI). Quotations may vary based on site assessments and customer negotiations.
5. **ROI (Return on Investment):** A metric used in quotations to show the financial benefits and cost efficiency of the solar installation to the customer. It helps customers assess the value of their investment.
6. **Rate:** The cost associated with the solar panels, installation, and other related services. Rates are configurable by the admin and can be predefined or entered as custom values by employees when creating a quotation.
7. **Performance Data:** Data related to an employee's sales performance, including the number of quotations created, the value of deals closed, and other sales metrics. This data is used for calculating commissions and analyzing employee productivity.
8. **Expense:** Costs associated with the solar installation, including solar panel costs, transportation, site analysis, tools, and other related expenses. These are included in quotations to provide an accurate total cost.

9. **Commission:** A financial incentive based on an employee's performance, typically tied to the number of deals closed or the value of the quotations. Admins can track and manage employee commissions based on their performance data.
10. **Web Interface:** The graphical user interface (GUI) that employees and admins interact with to access the Solar Quotation Management System. This interface is accessible via web browsers like Chrome, Firefox, etc.
11. **User Roles:** Categories of users within the system, including Admin and Employee roles, each with distinct levels of access and responsibilities in the system.
12. **Quotation Engine:** The system feature that allows employees to input customer and installation data, generate detailed quotations, and produce a report for the customer.
13. **Database:** A structured collection of data stored in the system, including customer information, employee data, quotations, rates, and expenses. The database allows efficient retrieval and management of this data.
14. **Configuration:** The settings and parameters of the system that can be adjusted by the admin, such as rate updates, user management, and system preferences.
15. **Site Assessment:** The process of evaluating a customer's installation site, including factors such as location, available space, and specific solar panel requirements. This assessment helps in creating accurate quotations.
16. **Negotiation:** The process of discussing and adjusting the terms of a quotation, including pricing, installation services, and other factors, to reach a final agreement with the customer.
17. **Quotation Validity:** The period during which the quotation is valid and applicable to the customer. After this period, a new quotation may need to be generated due to changes in rates, customer requirements, or other factors.
18. **Expense Tracking:** The process of managing and recording all expenses related to the solar installation, including materials, labor, and transportation costs, which are factored into the final quotation.

19. **Sales Report:** A report generated by the system to analyze employee performance, sales data, and commission calculations. Admins use this report to track performance and make decisions regarding employee incentives.

20. **System Configuration:** The setup and customization of the system by the admin, including user roles, rates, and other parameters that affect how employees interact with the system
